

Scenario

A cloud provider uses ONOS as its SDN controller. They want to expose network services (like virtual private networks or load balancers) to their tenants through a self-service portal.

Question: Which ONOS interface would be primarily used by this self-service portal to request and configure these network services?

Answer:

The self-service portal would primarily use the **Northbound Interface (NBI)** of the ONOS controller, specifically through the **ONOS REST API** (or higher-level **Intent Framework APIs**).

Explanation:

- **Northbound REST API:** Exposes high-level abstractions, configurations, and administrative functions to external management platforms, cloud orchestration frameworks (e.g., OpenStack, Kubernetes), and self-service web portals.
- **Intent Framework:** Allows the portal to submit declarative network requests (e.g., "connect Host A to Host B with X bandwidth" or "provision VPN endpoint") without needing to manage low-level flow rules or hardware details.